

Network Layer & Routing

Network Layer: Responsible for routing packets from source to destination across multiple networks.

IPv4

32-bit logical address, written in dotted-decimal notation. Provides 4.3 billion addresses.

IPv6

128-bit logical address, written in hexadecimal. Solves IPv4 exhaustion.

Routing

Determining the best path for a packet to reach its destination.

Distance Vector Routing:

Routers share knowledge of entire AS with only neighbors (e.g., RIP). Based on Bellman-Ford algorithm.

Link State Routing:

Routers share knowledge of neighbors with entire AS (e.g., OSPF). Based on Dijkstra's algorithm.

IP Addressing Classes:

- Class A: 1.0.0.0 to 126.0.0.0
- Class B: 128.0.0.0 to 191.255.0.0
- Class C: 192.0.0.0 to 223.255.255.0
- Class D: Multicast
- Class E: Experimental

Subnetting:

Dividing a large network into smaller, manageable sub-networks.